

# INSTAGRAM DATA ANALYSIS

## InternSpark Data Science Internship

### Student Details

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**Internship Role:** Data Science Intern

**Project Title:** Instagram Data Analysis

**Tools Used:** Python, Pandas, NumPy, Matplotlib, Google Colab

**Submission Date:** June 2026

## Introduction

Social media platforms play a major role in connecting businesses with their audiences. Among these platforms, Instagram is widely used for sharing content, building brand awareness, and increasing user engagement. Understanding how users interact with posts can help organizations improve their content strategy and reach a larger audience.

In this project, Instagram comment data was analyzed to study engagement patterns, posting activity, hashtag usage, and user interactions. The goal was to identify useful insights that can help improve engagement and create a more effective content strategy.

## Dataset Overview

The dataset contains Instagram comment-related information collected from multiple posts.

### Dataset Details

- Number of Records: 7,488
- Number of Columns: 8
- Unique Users: 77

- Unique Photos: 257

## Key Attributes

- Comment ID
- Comment Text
- User ID
- Photo ID
- Created Timestamp
- Posted Date
- Emoji Usage
- Hashtag Count

The dataset provides information about user engagement behavior and interaction patterns on Instagram.

# Data Cleaning and Preparation

Before starting the analysis, the dataset was cleaned and prepared using Python in Google Colab.

The following steps were performed:

- Loaded and inspected the dataset.
- Checked for missing values.
- Verified data types.
- Converted timestamp values into datetime format.
- Extracted hour and day information for posting schedule analysis.
- Confirmed that the dataset was ready for visualization and analysis.

|   | id | comment                | User id | Photo id | created Timestamp | posted date | emoji used | Hashtags used | count |
|---|----|------------------------|---------|----------|-------------------|-------------|------------|---------------|-------|
| 0 | 1  | unde at dolore         | 2       | 1        | 13-04-2023 08:04  | April 14    | yes        |               | 1     |
| 1 | 2  | quae ea ducimus        | 3       | 1        | 13-04-2023 08:04  | April 14    | no         |               | 2     |
| 2 | 3  | alias a voluptatum     | 5       | 1        | 13-04-2023 08:04  | April 14    | no         |               | 4     |
| 3 | 4  | facere suscipit sunt   | 14      | 1        | 13-04-2023 08:04  | April 14    | yes        |               | 2     |
| 4 | 5  | totam eligendi quaerat | 17      | 1        | 13-04-2023 08:04  | April 14    | yes        |               | 1     |

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7488 entries, 0 to 7487
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   id                    7488 non-null   int64
1   comment               7488 non-null   object
2   User id               7488 non-null   int64
3   Photo id              7488 non-null   int64
4   created Timestamp     7488 non-null   object
5   posted date           7488 non-null   object
6   emoji used            7488 non-null   object
7   Hashtags used count   7488 non-null   int64
dtypes: int64(4), object(4)
memory usage: 468.1+ KB
```

# Key Metrics

The following metrics were calculated from the dataset:

| Metric                | Value |
|-----------------------|-------|
| Total Comments        | 7,488 |
| Unique Users          | 77    |
| Unique Photos         | 257   |
| Average Hashtags Used | 2.42  |

These metrics provide a general overview of user engagement within the dataset.

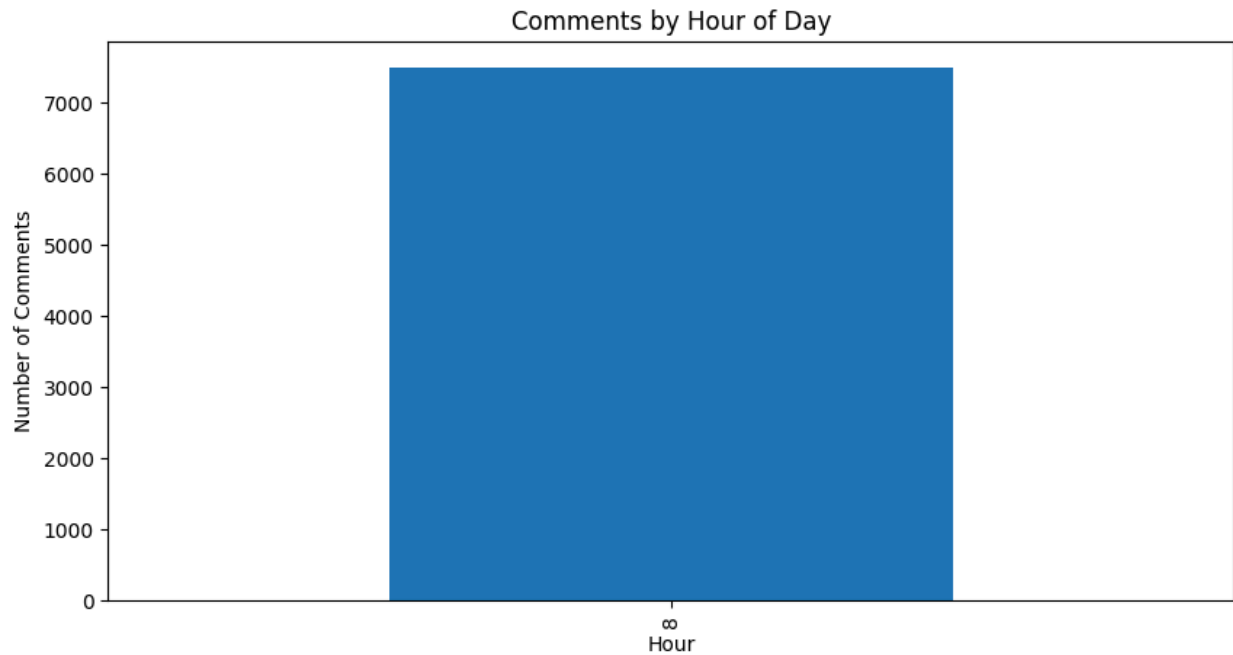
```
Total Comments: 7488
Unique Users: 77
Unique Photos: 257
Average Hashtags Used: 2.42
```

# Exploratory Data Analysis

## Posting Time Analysis

Comment activity was analyzed across different hours of the day to identify periods of higher engagement.

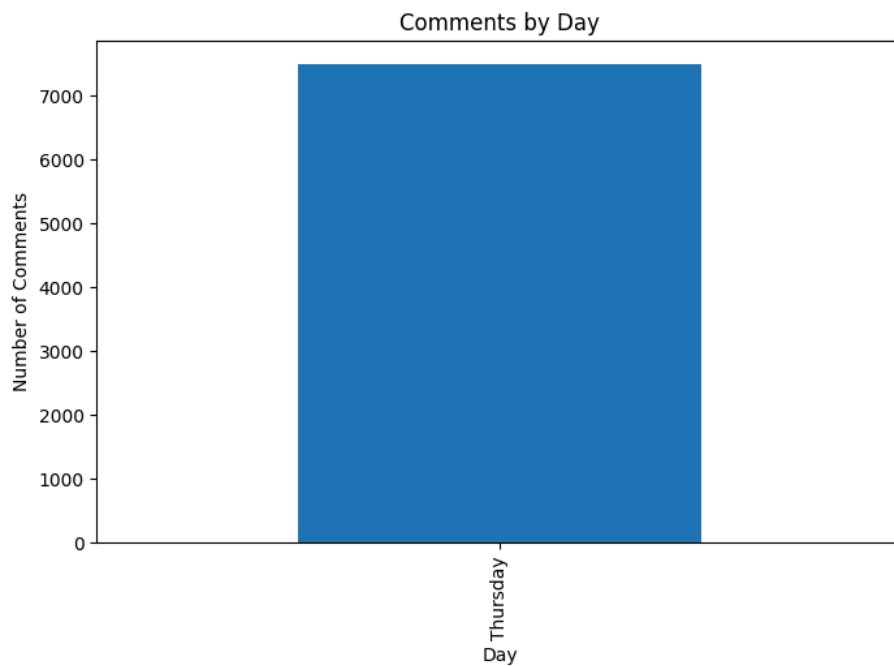
The analysis helps determine the best times to publish content and interact with audiences.



## Day-wise Engagement Analysis

Comment activity was also examined across different days of the week.

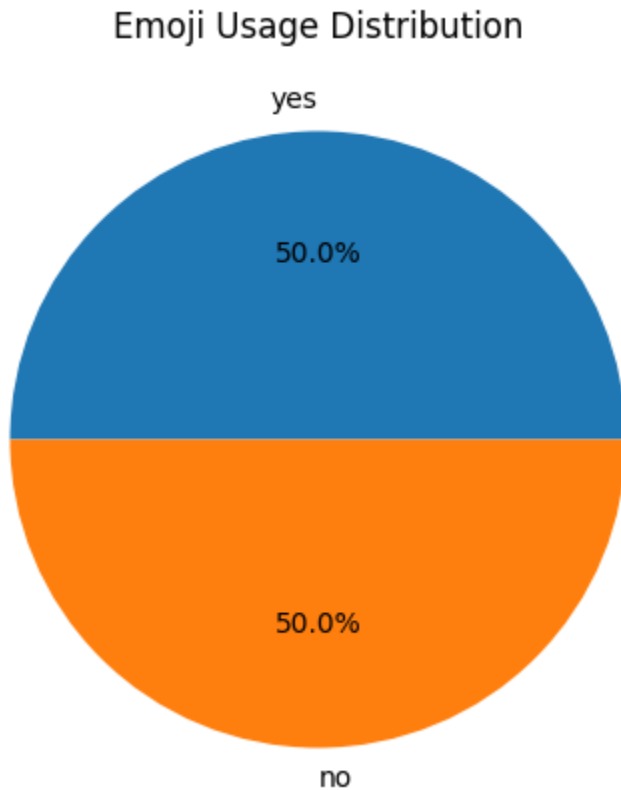
This analysis helps identify days that generate stronger audience participation.



## Emoji Usage Analysis

Emoji usage was analyzed to understand how users express reactions and emotions within comments.

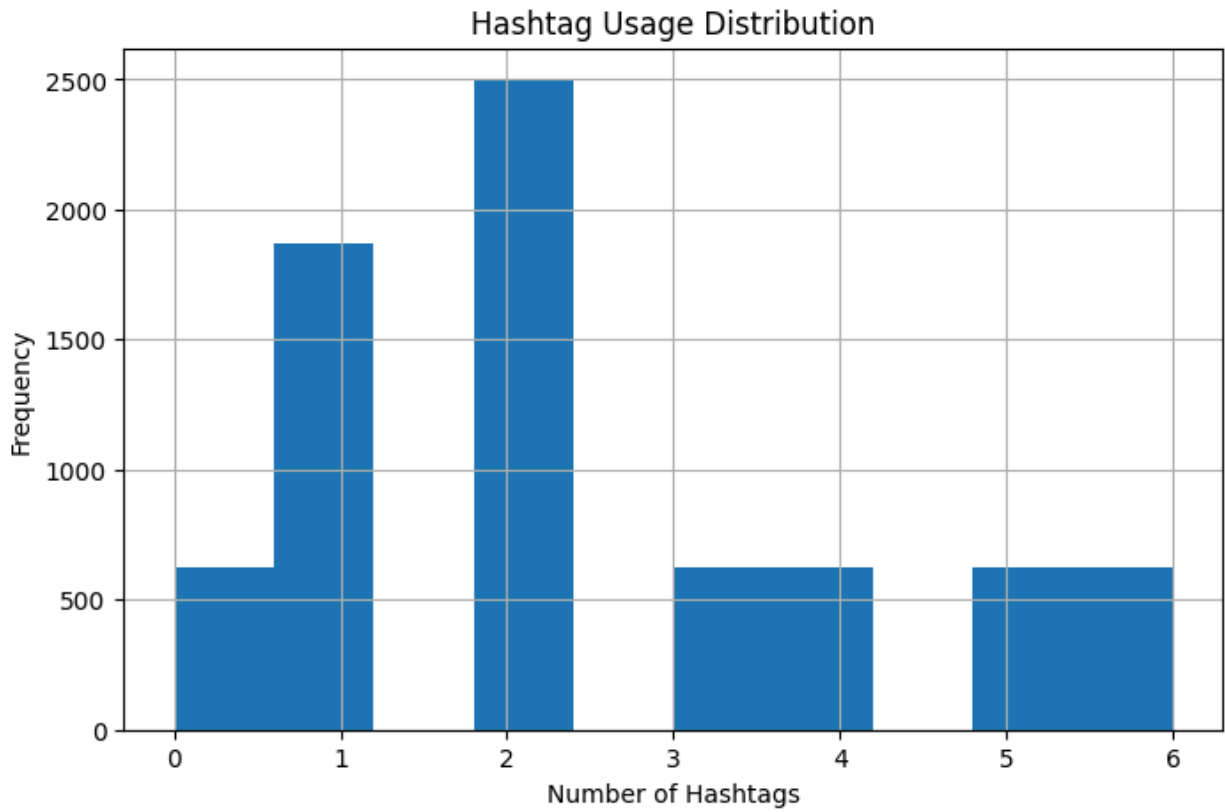
The results indicate the frequency of emoji usage among Instagram users.



## Hashtag Analysis

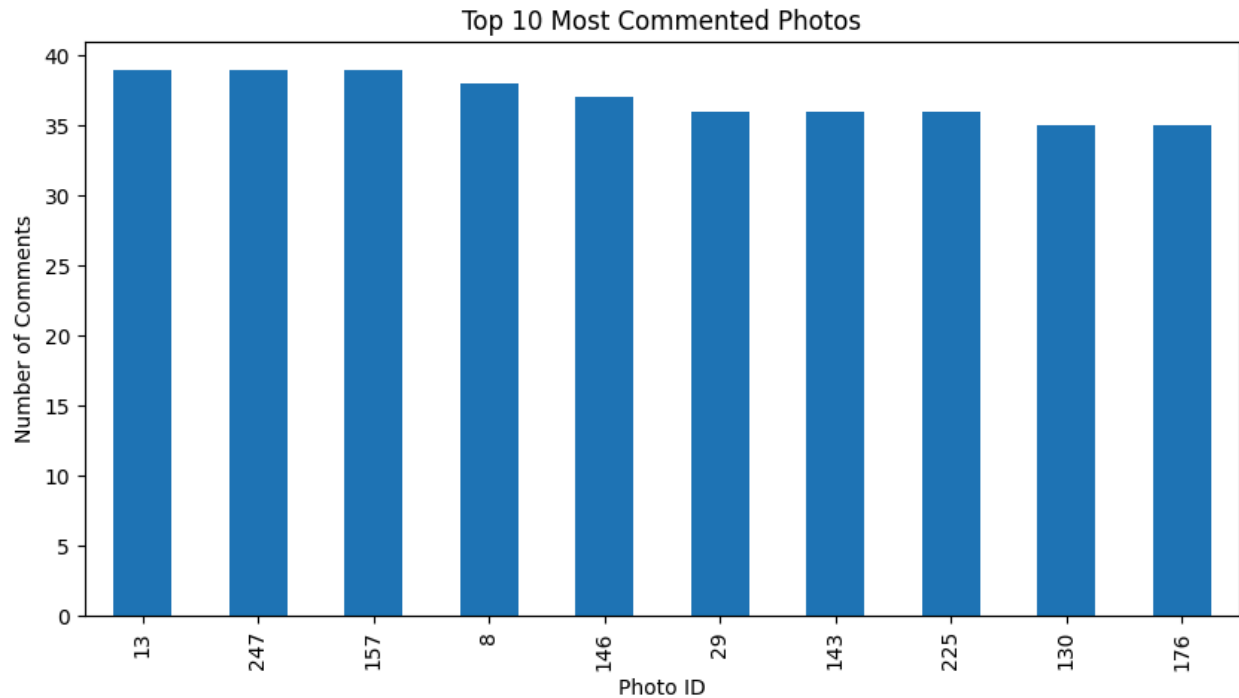
Hashtags play an important role in increasing content visibility and discoverability.

The analysis examined the number of hashtags used and their overall distribution.



## Most Engaged Posts

The most commented photos were identified to understand which posts generated the highest level of audience interaction.



## Key Findings

1. The dataset contains 7,488 comments across 257 Instagram posts.
2. A total of 77 unique users actively interacted with content.
3. User engagement varied throughout the day, suggesting that posting time influences interaction levels.
4. Certain days generated higher comment activity than others.
5. Emoji usage was common among users and contributed to engagement.
6. The average number of hashtags used per interaction was approximately 2.42.
7. Some posts attracted significantly more comments than others, indicating higher audience interest.
8. Consistent posting and strategic hashtag usage can improve engagement performance.

## Recommendations for Alfido Tech

Based on the analysis, the following recommendations are suggested:

1. Publish content during peak engagement hours identified through comment activity analysis.
2. Use relevant hashtags consistently to improve content visibility and reach.
3. Encourage audience participation through polls, questions, and discussion-based posts.

4. Include emojis where appropriate to create more engaging and relatable content.
5. Maintain a regular posting schedule to keep followers active and engaged.

# One-Week Content Strategy Plan

## Monday

Educational posts, tutorials, and industry insights.

## Tuesday

Interactive posts such as polls and audience questions.

## Wednesday

Success stories, project highlights, and achievements.

## Thursday

Behind-the-scenes content and team activities.

## Friday

Trending technology topics and industry updates.

## Saturday

Community engagement posts and user-generated content.

## Sunday

Weekly recap and upcoming announcements.

# Executive Summary

This project focused on analyzing Instagram engagement data using comment activity, posting schedules, emoji usage, and hashtag patterns. The dataset contained 7,488 comments generated by 77 users across 257 posts. After preprocessing and cleaning the data, multiple visualizations were created to explore user interaction trends.



The analysis showed that engagement levels vary depending on posting time, content interaction, and hashtag usage. Certain posts generated significantly higher comment activity, indicating stronger audience interest. Emoji usage and hashtags were also found to be important factors in encouraging user interaction.

Based on the findings, several recommendations were developed to help improve engagement and strengthen content performance. The project demonstrates how data analysis can support better social media decision-making and content planning.

## Conclusion

The Instagram Data Analysis project provided useful insights into audience engagement patterns and social media interaction behavior. By analyzing comments, posting schedules, hashtag usage, and user activity, meaningful trends were identified that can help improve content performance.

The findings suggest that consistent posting, effective hashtag strategies, and audience-focused content can significantly improve engagement levels. These insights can help organizations such as Alfido Tech develop stronger social media strategies and build better relationships with their online communities.

Overall, this project highlights the importance of data-driven decision-making in social media marketing and audience engagement.